Homework 1: Image Enhancement

Computer Vision - Fall 2026

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1 Objective

This project compares classical image enhancement techniques applied to color images artificially degraded by reduced contrast and additive Gaussian noise. The methods are implemented using OpenCV and NumPy, without relying on neural networks.

The objective is to evaluate their ability to improve image quality by considering contrast enhancement, noise reduction, and detail preservation. Both quantitative metrics and visual comparisons are used to assess the results and identify the trade-offs between numerical performance and perceptual quality.

2 Images and Degradation

Three color images were used in this project: `parigi.png`, `cavallo.png`, and `mare.png`. All three images are original photographs taken by the author. They were selected to represent different visual characteristics, including architectural structures, natural textures, and non-uniform illumination.

Each original image f was synthetically degraded by reducing its contrast and adding Gaussian noise. The resulting image g is defined as:

$$g = \text{clip}\left(\alpha f + \beta + n,\, 0,\, 255\right), \qquad n \sim \mathcal{N}(0, \sigma^2)$$

The degradation parameters were fixed to $\alpha = 0.6$, $\beta = 0$, and $\sigma = 15$. A fixed random seed (`seed=42`) was used to ensure reproducibility. The degradation was implemented using `cv2.convertScaleAbs` and NumPy.

The original images were used as clean reference images for quantitative evaluation. However, in realworld scenarios, the original clean image is often unavailable.

3 Methods

All methods were applied to the same degraded images using fixed parameters.

1. **Baseline:** the degraded image without any additional processing.
2. **Linear Contrast Stretching:**
   - **2A − Luminance:** min-max normalization applied only to the L channel in the CIE-Lab color space, leaving the chromatic channels a and b unchanged.
   - **2B − Per-channel:** independent normalization of the B, G, and R channels, which may alter color relationships and introduce color shifts.
3. **Histogram Equalization (3):** applied to the L channel in the CIE-Lab color space using `cv2.equalizeHist`. It improves contrast but can also make noise more visible in plain areas.
4. **Denoising + Contrast Stretching (4):** noise reduction followed by contrast enhancement on the luminance channel:
   - **4A:** bilateral filter (`d=10`, `sigmaColor=80`, `sigmaSpace=80`).
   - **4B:** Non-Local Means (`h=13`, `hColor=13`, `templateWindowSize=7`, `searchWindowSize=21`).

Denoising reduces noise but can also remove small details. Therefore, it is important to compare how well each method removes noise while keeping image details.

4 Metrics

The results were compared with the original images using three metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), and Peak Signal-to-Noise Ratio (PSNR).

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(f_i - \hat{f}_i)^2, \qquad \text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|f_i - \hat{f}_i|.$$

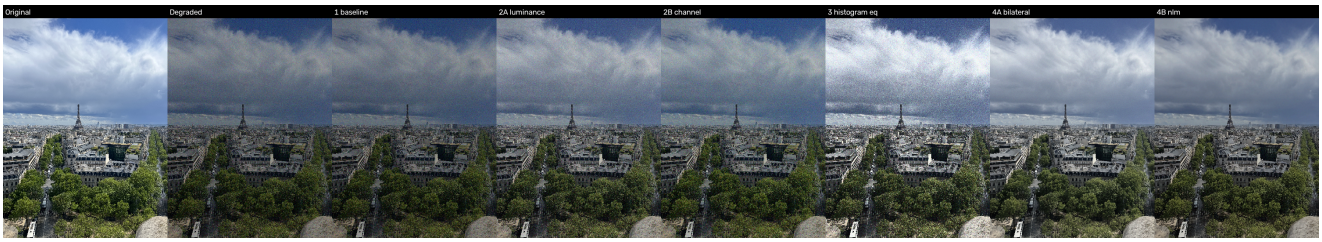
$$\text{PSNR} = 10\log_{10}\left(\frac{255^2}{\text{MSE}}\right) \text{ dB}.$$

MSE and MAE measure the difference between the original and enhanced images. Lower values indicate smaller differences. PSNR measures image quality: higher values indicate better results.

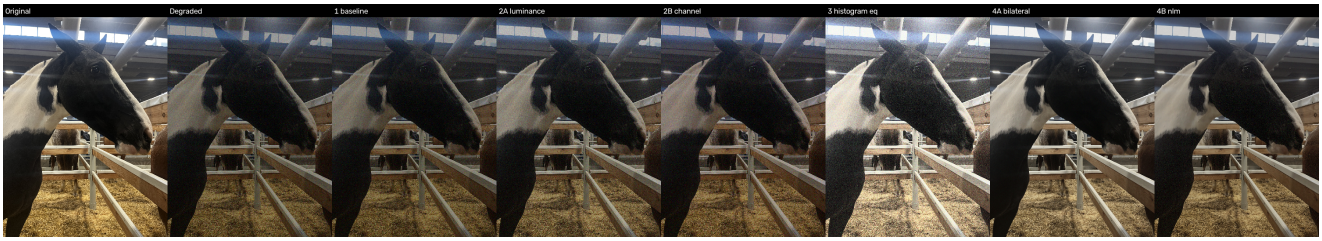
5 Results

5.1 Visual Comparison

Figure 1 shows the original images, their degraded versions, and the results obtained with the six methods.



(a) Paris



(b) Cavallo

(c) Mare

Figure 1: Visual comparison of the three images. From left to right: original, degraded, baseline (no processing), 2A (luminance stretching), 2B (per-channel stretching), 3 (histogram equalization), 4A (bilateral denoising + stretching), and 4B (Non-Local Means + stretching).

5.2 Quantitative Results

Table 1 reports the PSNR and MAE values for each method and each image. The best values are shown in bold. MSE values are available in `results/metrics.csv`.

	Parigi		Cavallo		Mare	
Metodo	PSNR ↑	MAE ↓	PSNR ↑	MAE ↓	PSNR ↑	MAE ↓
1 Baseline	12.42	53.03	14.37	39.55	12.59	51.75
2A Stretching (L)	15.93	34.40	16.56	30.52	16.45	32.31
2B Stretching (BGR)	14.82	39.45	15.81	33.36	15.17	36.91
3 Equalizzazione (L)	19.65	20.94	17.13	28.75	20.00	20.15
4A Bilateral + stretch	**22.89**	**14.96**	**24.87**	**11.68**	**28.12**	**7.25**
4B NLM + stretch	17.28	30.63	19.86	21.89	24.14	13.92

Table 1: PSNR (dB) and MAE values for each method.

The method 4A achieves the best numerical results for all three images. Method 2A performs better than 2B. Method 4B performs better than histogram equalization for the Cavallo and Mare images, but not for Paris.

6 Discussion

The results show that the best numerical performance does not always mean the best visual quality.

Method 4A achieves the highest PSNR and the lowest MAE for all three images. However, the bilateral filter can make textures too smooth and remove small details. Method 4B produces visually similar results, although some noise remains.

Visual Analysis.
For the horse image, method 4A gives the best visual result because the details of the fur are more visible.
For Paris, method 4B looks better because the details of the buildings are more visible.
The sea image is more difficult to evaluate because the waves make the comparison less clear. We can see that the method 4B produces colors closer to the original image, but the result is still too dark.

Other methods also have some limitations, for example histogram equalization can make noise more visible in plain areas, and per-channel stretching may change the original colors.

Therefore, even if PSNR, MSE, and MAE show better numerical results they do not always reflect visual quality. An image may still lose small details and textures. For this reason, numerical results should always be compared with the images.

(a) 4A: Bilateral filtering + stretching

(b) 4B: Non-Local Means + stretching

Figure 2: Comparison between methods 4A and 4B on the Parigi image.

As shown in Figure 2, a method with better numerical results does not always produce the best visual result. Although 4A has better metrics, 4B makes the building details more visible.

7 Limitations and Conclusions

The synthetic degradation does not fully represent real-world conditions. The experiment was carried out using only three images, with the same parameters for all of them.

I also decided to test method 2B (per-channel contrast stretching) to see how it differs visually from method 2A (luminance contrast stretching). I noticed that 2B changes the original colors, while 2A preserves them better.

I also tested method 4B (Non-Local Means) because method 4A made some areas of the images too smooth. I wanted to compare it with another denoising method that could reduce noise while keeping more details, instead of removing details that I considered important.

In conclusion, none of the methods can fully restore the original image because some information is lost during the degradation and enhancement processes.

8 Reproducibility

The source code, original and degraded images, enhanced results, and `metrics.csv` are available in the repository: `https://github.com/yezmiv/Computer-Vision-homeworks/tree/main/HW1`.

The project uses Python 3.14.3, OpenCV, and NumPy. The complete pipeline can be run with the following command:

```
python3 homework1.py
```